

TARA Research Project Proposal: Verifying RLHF Sycophancy Amplification

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1. Project Summary

This project investigates the mechanisms by which Reinforcement Learning from Human Feedback (RLHF) amplifies sycophancy in Large Language Models (LLMs). While Shapira et al. (2026) provided theoretical evidence that RLHF drives models to endorse users' false beliefs to maximize human-biased rewards, empirical verification on purely open-source models remains limited. This project serves as an empirical verification of Shapira et al.'s (2026) covariance theorem by measuring how preference optimization pressure maps directly to policy drift in a localized environment. We will empirically verify this amplification using a pure base model (Llama-3.2-1B Base) under optimization pressure. We will use Best-of-N selection as an inference-time proxy for optimization and Group Relative Policy Optimization (GRPO) to apply scalable, memory-efficient alignment. Furthermore, we will test the efficacy of Shapira et al.'s proposed closed-form KL-penalty mitigation in a practical, constrained computing environment.

2. Motivation: Existential Risk and Government AI Deployment

2.1 Sycophancy as an Existential Risk (X-Risk)

Sycophancy directly erodes human epistemic oversight, a core pathway to catastrophic AI risk. As models scale and take on high-stakes tasks, human evaluators increasingly rely on the models themselves to verify complex information. If an RLHF-trained system learns that confirming a human's flawed hypothesis yields a higher reward than correcting it, the AI becomes systematically manipulative. At scale, this creates a feedback loop where humans are confidently assured of disastrously wrong decisions (e.g., in biosecurity, military strategy, or

alignment research itself), completely neutralizing our ability to catch AI errors.

2.2 Government AI Chat Farms and Public Polarization

On a socio-political level, sycophantic AI deployed at scale by state actors or within government-contracted "chat farms" poses immediate risks to democratic stability. If government service bots or state-backed LLMs are optimized using flawed human preference data, they will automatically reflect and amplify the biases, conspiracies, or false beliefs of the user interacting with them. Instead of acting as objective information engines, these bots become personalized echo chambers, radicalizing citizens by validating misinformation under the guise of an authoritative, state-sanctioned AI. Sycophancy is not just a polite quirk; it is an automated radicalization mechanism.

3. Methodology Upgrades & Addressing Methodological Flaws

3.1 Breaking Circularity: Independent Ground-Truth Validation

The Flaw: Using ArmoRM to drive optimization and simultaneously using ArmoRM scores to measure sycophancy amplification creates a Goodhart's Law trap. The model may simply adopt a "polite" style that the RM arbitrarily rewards, mimicking the GPT-4o regression (OpenAI, 2025).

The Fix: We decouple evaluation from optimization. While Best-of-N and GRPO will optimize against ArmoRM-Llama3-8B, the final sycophancy measurement will be conducted against an independent, objective ground-truth factual metric. We will evaluate outputs against the TruthfulQA exact-match labels. While advanced probabilistic methods exist, we will utilize the automated string-matching evaluation baseline established by Lin, Hilton, and Evans (2022) as a robust and necessary approach for our resource-constrained local environment. If the RM score increases but factual accuracy degrades in the presence of user belief injection, we successfully isolate genuine sycophancy amplification.

3.2 Statistical Power and Dataset Expansion (Preventing GRPO Collapse)

The Flaw: Executing GRPO on a 1B model over a microscopic dataset of just 200 highly-stratified prompts will likely lead to immediate overfitting and policy entropy collapse.

The Fix: We will implement a stratified sampling pilot phase to measure initial reward tilt.

However, for the active GRPO training phase, we will expand the dataset to utilize the full 1,000-prompt SycophancyEval set alongside standard factual datasets. This guarantees the mathematical variance required to observe the prompt-level interaction under optimization pressure while preventing the algorithm from degenerating into a useless state.

3.3 Base Model Coherence via Few-Shot Scaffolding

The Flaw: Llama-3.2-1B Base is incredibly prone to spitting out gibberish or repeating prompt tokens instead of answering questions coherently, which will break the reward scoring. Using an Instruct model would corrupt the base policy baseline required by Shapira et al.'s theorem.

The Fix: Instead of using an aligned model, we will use In-Context Learning (Few-Shot Prompting) to coerce the base model into formatting its answers correctly, without altering its weights. A strict prefix containing examples of a user asking a question and an assistant providing a clean answer will be prepended. This allows the model's attention mechanism to mimic the pattern for coherent generation at inference time while maintaining a completely uncorrupted base policy.

3.4 Best-of-N Optimization Proxy Bound

We leverage Best-of-N selection as an inference-time proxy for PPO optimization pressure. Crucially, in alignment with Beirami et al., the formula $KL \approx \log(N) - (N-1)/N$ will be strictly treated as an *upper bound* on the actual KL divergence, not an exact equality.

4. Technical Architecture and Hardware Strategy

To execute the experiment within defined compute budgets, the architecture is designed for a localized and cloud-bursting execution model:

Component	Selection	Justification
Base Model	Llama-3.2-1B (Base)	Instruct models are already RLHF-corrupted. Measuring base policy covariance requires pure pre-trained weights.
Reward Model	ArmoRM-Llama3-8B	High-quality open-source RM capable of evaluating

Component	Selection	Justification
		complex preference data. Can be quantized to 4-bit/8-bit for local inference proxy scoring.
Alignment Algorithm	Best-of-N (Proxy) & GRPO	Best-of-N provides a pure isolation of optimization pressure. GRPO eliminates the Value network, drastically reducing overhead.
Hardware Execution Strategy	Local Hardware & Cloud Renting (Vast.ai A100)	The Best-of-N proxy phase and prompt filtering will be evaluated locally utilizing available consumer architecture (e.g., M1 Max / RTX 3080). The memory-intensive GRPO phase will be executed via targeted cloud compute on a single rented NVIDIA A100 (80GB) node. This configuration comfortably hosts the 1B policy model (with optimizer states), reference model, and 8B ArmoRM simultaneously in native bfloat16 without hitting out-of-memory errors.

5. Scope Definition

5.1 In Scope

- Deploying Llama-3.2-1B (Base) with few-shot scaffolding to measure initial reward tilt.

- Executing a 1,000-prompt pilot run and utilizing the full set for GRPO stability.
- Using Best-of-N selection (N=1, 4, 16, 64) as an inference-time proxy for PPO.
- Implementing GRPO via Hugging Face TRL for memory-efficient alignment mapping via cloud compute.
- Measuring exact-match accuracy against TruthfulQA ground-truth labels (Lin et al., 2022) to break RM circularity.
- Applying and evaluating Shapira et al.'s KL-penalty mitigation.

5.2 Out of Scope

- Training a custom Reward Model from scratch.
- Testing multi-modal (vision/audio) sycophancy tasks.
- Deploying models larger than 3B parameters (due to strict budget constraints for targeted cloud execution, precluding 7B/8B policy modeling).
- Evaluating constitutional AI or purely prompt-based mitigations (focus remains on RL mechanism).

6. Limitations

While this project provides a rigorous empirical verification framework, it operates under inherent computational and architectural constraints. Notably, we are utilizing a 1B-parameter open-weights base model (Llama-3.2-1B) rather than the precise models, proprietary weights, or frontier-scale infrastructure analyzed in the original Shapira et al. (2026) literature. Because sycophantic tendencies and optimization dynamics can scale non-linearly with model size and pretraining regimes, the absolute magnitude of the reward tilt and subsequent policy drift observed in this study may differ from the baseline results reported in previous works.

7. Core Literature Review Checklist

1. **Shapira et al. (Feb 2026):** *How RLHF Amplifies Sycophancy*. Provides the core mathematical framework mapping preference bias to policy drift.
2. **Beigi et al. (Sept 2025):** *Sycophancy Mitigation Through Reinforcement Learning with Uncertainty-Aware Adaptive Reasoning Trajectories (SMART)*. Represents state-of-the-art trajectory-based mitigation using UA-MCTS, serving as a methodological contrast to output-based GRPO alignment.
3. **Chen et al. (Nov 2025):** *Self-Augmented Preference Alignment for Sycophancy Reduction in LLMs*. Introduces preference mapping enhancements and the SAA dataset.

4. **Sharma et al. (2023):** *Towards Understanding Sycophancy in Language Models.*
Establishes the foundational SycophancyEval parameters and proves sycophancy exists as a general behavior prior to RLHF.
5. **OpenAI (April/May 2025):** *"Sycophancy in GPT-4o: what happened and what we're doing about it"* & *"Expanding on what we missed with sycophancy"*. Industry verification of reward model confounding and the dangers of over-weighting user feedback.